Revisiting Chart-Based Return Prediction: Shared Rankings and Economic Limits

Abstract

We replicate and extend the controlled representation comparison of Jiang et al. through 2024. Holding recent price and volume histories fixed, we compare logistic classifiers, one-dimensional convolutional neural networks (CNN1D), and a chart CNN under alternative scaling schemes. After local high–low scaling, even the logistic classifier produces a strong weekly return ranking, and CNN1D applied to the continuous sequence reproduces the chart CNN's weekly portfolio performance. On exact common stock-date panels, chart and sequence scores rank many of the same stocks but are not identical; feature-level and prediction-level fusion add little to the strongest individual model. The economic value is narrower than the unrestricted results suggest. Learned portfolios remain strong under a common transaction-cost schedule, but net performance becomes negative under value weighting and modest tradability screens. Learned models and short-horizon price rules also weaken during 2020–2024. The representation result therefore persists, while the score comparisons reveal largely shared predictive content and the implementation tests show limited scalable value.

Keywords: Return predictability, Machine learning, Representation learning, Market liquidity, Transaction costs

JEL classification: C45, G11, G12, G14

1. Introduction

Chart-based models raise a basic question for return prediction: when a machine learns from a market chart, what component of the prediction pipeline produces the ranking? A chart is not raw market history. Before a model sees it, the input window has been selected, prices have been scaled within that window, and price and volume information has been rendered in a fixed visual layout. A flexible model is then applied to the resulting representation. The strong portfolio performance of chart convolutional neural networks (chart CNNs) (Jiang et al. 2023) may therefore reflect the chart image, the local high–low scaling embedded in chart construction, or nonlinear learning from recent price-volume information.

Recent market history can be represented through prespecified price rules, ordered numerical sequences, or chart images. Lo et al. (2000) show how chart patterns can be defined and tested systematically, while Murray et al. (2024) study nonlinear predictability in numerical return histories. Jiang et al. (2023) provide the direct starting point for our analysis. Beyond applying a CNN to price and volume images, they compare logistic classifiers and CNN1D across image, cumulative-return, and devolatized-return scaling for all nine input-window and forecast-horizon combinations under equal and value weighting. They find that CNN1D under the image-scale specification roughly matches or exceeds the chart CNN and identify local high–low scaling as a key component of the image representation. The representation comparison itself is therefore not new. The unresolved questions are whether similar portfolio performance reflects the same stock rankings, whether combining chart and sequence representations improves prediction, and whether the economic results persist in recent data and more tradable universes.

We replicate and extend this design. We first re-estimate the logistic, CNN1D, and chart-CNN comparison through 2024. We then move beyond portfolio-level performance by forming exact common stock-date panels for the learned-model scores. Rank correlations and joint encompassing regressions measure overlap and conditional predictive content, while feature-level and prediction-level fusion test whether the chart and sequence representations contain complementary information. Model development remains confined to 1993–2000, and all portfolio results are evaluated out of sample over 2001–2024 under common ranking, weighting, holding-period, and transaction-cost rules.

The replication confirms Jiang et al.'s representation result in the extended sample. In the representative $I20/R5$ specification, the image-scale logistic classifier generates an equal-weighted gross Sharpe ratio of 5.22, compared with -0.41 under cumulative-return scaling. CNN1D raises the Sharpe ratio to 6.55, close to 6.52 for the chart CNN, and remains competitive across the other weekly input windows. The score-level tests add a qualification that portfolio Sharpe ratios cannot reveal: the model rankings overlap substantially but are not identical. Neither feature-level nor prediction-level fusion consistently improves on the strongest individual representation. The evidence therefore confirms the importance of local high–low scaling while assigning limited incremental portfolio performance to the binary chart layout.

The economic analysis likewise reassesses Jiang et al.'s conclusions using the same four price-based benchmarks and broad transaction-cost schedule. Full-universe equal-weighted portfolios retain net Sharpe ratios near four, consistent with their finding that stated costs do not eliminate portfolio performance. The conclusion changes under modest dollar-volume and market-capitalization screens, where net returns become negative, and under value weighting. The chart CNN, CNN1D, multimodal model, and short-horizon price rules also weaken during 2020–2024. These results narrow the original economic interpretation: historical performance is concentrated in equal-weighted portfolios and stocks that are difficult to trade at scale, and its magnitude attenuates after 2019. This boundary is consistent with recent evidence on implementation constraints and decay in chart-CNN performance (Kaczmarek and Pukthuanthong 2025).

The contribution lies in moving from a portfolio-level replication to direct tests of representation overlap, complementarity, persistence, and economic scope. Exact score comparisons show what similar Sharpe ratios conceal, fusion tests ask whether the representations improve one another, and the extended sample and tradability screens revise the assessment of scalable value. Section 2 presents the replication design and extensions. Section 3 reports the representation comparison and score-level tests. Section 4 reassesses benchmark performance, implementation, and post-2019 stability. Section 5 concludes.

2. Data, inputs, and empirical design

The prediction pipeline combines local high–low scaling and a binary chart representation with flexible learning. Building on the controlled representation comparison in Jiang et al. (2023), we first reproduce the logistic, CNN1D, and chart-CNN comparisons under alternative scaling schemes. The added tests then place the learned-model scores on exact common stock-date panels and examine ranking overlap, conditional predictive content, and multimodal complementarity.

Timing, ranking, weighting, and transaction-cost rules remain fixed across the learned-model evaluations.

2.1. *Sample and prediction target*

We construct a daily CRSP panel of ordinary common shares listed on NYSE, AMEX, or Nasdaq using share codes 10 and 11. The panel includes daily returns, open, high, low, and close prices, trading volume, lagged market capitalization, stock identifiers, and available delisting returns. Eligibility requires positive prices and sufficient history for the relevant specification. Firms enter and leave with their CRSP histories, producing an unbalanced panel without a survivorship requirement.

Model development uses data from 1993–2000, and portfolio evaluation is out of sample from 2001 through 2024. We report the full evaluation period, a 2001–2019 baseline sample, and a 2020–2024 extension sample. The model grid combines input windows $I \in \{5, 20, 60\}$ with future return horizons $R \in \{5, 20, 60\}$. We write a 20-day input and 5-day target as $I20/R5$. For stock i at formation date t, the directional target is

$$y_{i,t}^{(R)} = \mathbb{1}\left(\prod_{j=1}^{R}(1 + r_{i,t+j}) - 1 > 0\right). \tag{1}$$

The target begins at $t + 1$, after all input information has been observed by the close of date t.

To remove mechanical price jumps from stock splits, we set the first close in each input window to one, reconstruct subsequent closes from daily returns, and scale open, high, and low prices consistently with the reconstructed close series. The moving-average window equals I. Within a specification, the learned models use the same underlying observations. Eligible cross-sections can differ across input windows and price-based benchmarks because their history requirements differ. Portfolio results use the sample available to each score, while direct learned-model overlap tests impose an exact common stock-date panel.

2.2. *Input construction and representations*

The chart input combines local high–low scaling with a binary grayscale representation. We use *local high–low scaling* for this transformation in the prose and *image scale* in specification labels and tables. For each stock and formation date, let $m_{i,t}^{(I)}$ and $M_{i,t}^{(I)}$ denote the minimum and maximum calculated jointly across all open, high, low, close, and moving-average observations in the lookback window. Each price-channel observation is transformed as

$$\widetilde{x}_{i,t-j} = \frac{x_{i,t-j} - m_{i,t}^{(I)}}{M_{i,t}^{(I)} - m_{i,t}^{(I)}}. \tag{2}$$

Trading volume is scaled separately within the same window. Chart construction then places daily OHLC bars and a moving-average line in the price panel and volume in a lower panel. The 5-, 20-, and 60-day images contain 32×15, 64×60, and 96×180 binary pixels, respectively. The online appendix provides an input illustration and the complete representation map.

The numerical sequence contains the same six daily channels in chronological order: open, high, low, close, moving average, and volume. We evaluate three sequence specifications. The image scale retains the continuous within-window values from Eq. (2); the cumulative-return scale anchors each price channel to the first close; and the devolatized-return scale expresses price changes relative to lagged exponentially weighted volatility and volume through relative changes. Each transformed channel is standardized using development-sample moments that remain fixed during evaluation.

The chart and locally scaled sequence therefore share the same local high–low scaling but differ in representation. The sequence retains continuous observations ordered by trading day, whereas the chart converts them into a fixed binary pixel layout. Logistic comparisons across scaling schemes measure how the input transformation changes a linear ranking. Comparing the logistic classifier with CNN1D under the image-scale specification shows how the ranking changes when nonlinear temporal learning is added, while comparing CNN1D with the chart CNN asks whether the binary image is necessary to attain similar portfolio performance.

This comparison narrows the source of performance without creating a pure treatment for image layout. The chart CNN and CNN1D also differ in numerical precision, dimensionality, filter geometry, depth, and capacity. Similar performance after local high–low scaling therefore indicates that the tested portfolio pattern can be recovered without the binary chart image; residual differences can reflect any of the remaining representation or architecture features.

2.3. Prediction models and estimation

The chart CNN follows the design of Jiang et al. (2023) and applies two-dimensional convolutions to the binary images. CNN1D applies one-dimensional convolutions along the trading-day dimension of the six-channel sequence. Both architectures increase in depth with the input window. The logistic specifications flatten the sequence into a linear index. The online appendix reports the complete architecture, scaling, and training settings.

The multimodal model pairs chart and sequence inputs for the same stock, formation date, input window, and forecast horizon. Separate chart CNN and CNN1D branches produce feature vectors that are concatenated before a linear classification head. We also examine late fusion by averaging the predictions of separately trained branches. Each model's portfolio score is the softmax probability assigned to the positive-return class; it is used for within-date ranking and is not interpreted as a calibrated expected return.

We estimate separate models across input windows, forecast horizons, model families, sequence scales, and fusion designs. Neural models minimize cross-entropy loss with Adam and stop when development-sample validation loss fails to improve for two consecutive epochs. Stock-date observations from 1993–2000 are randomly divided into 70% training and 30% validation sets using a fixed seed, and the lowest-validation-loss checkpoint remains fixed throughout the 2001–2024 evaluation period. The main analysis uses one selected checkpoint per specification. The appendix reports all estimation settings, a five-member chart-CNN sensitivity check for the $I5/R5$ and $I20/R5$ specifications, and a compact comparison of verified design similarities and differences relative to Jiang et al.

2.4. Portfolio construction and empirical tests

At each formation date, the eligible cross-section is sorted into ten deciles using the relevant model score or benchmark signal. The high-minus-low portfolio buys the highest-score decile and shorts the lowest-score decile. Equal-weighted portfolios assign the same weight to every stock within a leg; value-weighted portfolios use lagged market capitalization normalized separately within each leg. Each leg carries one dollar of notional, so the spread has zero net investment and two dollars of gross exposure.

Weekly, monthly, and quarterly portfolios correspond to the 5-, 20-, and 60-trading-day forecast horizons. Information through the close of date t ranks stocks for the return from $t + 1$

through $t + R$. One formation date per holding interval produces non-overlapping return series. For a portfolio return series $R_{p,t}$ with $K \in \{52, 12, 4\}$ observations per year, the annualized Sharpe ratio is

$$\mathrm{SR}_p = \sqrt{K}\,\frac{\overline{R}_p}{\mathrm{sd}(R_{p,t})}. \tag{3}$$

No risk-free rate is subtracted from the zero-net-investment spread.

We report the full grid across input windows and forecast horizons. The later economic tests use $I20/R5$ as the representative weekly specification because it is central to the short-horizon grid and performs strongly across all three learned model families. This specification was selected after examining the grid, so the focused comparisons are descriptive. Benchmark signals can require longer histories than the learned models, and standalone portfolios therefore need not contain identical stock-date observations.

Price-based benchmarks.. We compare the learned rankings with momentum (MOM), one-month short-term reversal (STR), one-week short-term reversal (WSTR), and the multi-horizon trend strategy (TREND). MOM is a daily analogue of 12–1 momentum (Jegadeesh and Titman 1993); STR and WSTR use the negative of the previous 21- and 5-trading-day returns (Jegadeesh 1990; Lehmann 1990); and TREND combines moving-average-to-price ratios across horizons using lagged forecasting coefficients (Han et al. 2016). All benchmarks follow the common portfolio protocol. The appendix reports their complete formulas and result grids.

Costs and implementation.. Net performance applies one-way costs to full absolute traded notional in both legs: 10 basis points for stocks above the contemporaneous NYSE 80th-percentile market-capitalization breakpoint and 20 basis points otherwise. Reported turnover is conventional half-turnover summed across the two legs and expressed as a monthly equivalent. The cost schedule excludes stock-specific spreads, market impact, borrowing costs, short-sale constraints, and capacity limits. We also re-form the representative portfolios after price, dollar-volume, and NYSE market-capitalization screens while leaving model scores fixed. Carhart four-factor regressions use weekly net high-minus-low returns, annualized intercepts, and Newey–West standard errors with four lags (Carhart 1997).

Ranking overlap and conditional content.. For each input window and forecast horizon, we form an exact common panel containing all three learned-model scores and the future return. Date-level Spearman correlations measure similarity in the within-date rankings (Spearman 1904). We also estimate the cross-sectional regression

$$R_{i,t+1:t+R} = \alpha_t + \beta_t^{2D} s_{i,t}^{2D} + \beta_t^{1D} s_{i,t}^{1D} + \beta_t^{MM} s_{i,t}^{MM} + \varepsilon_{i,t+R}. \tag{4}$$

The Fama–MacBeth coefficients average the date-level slopes through time (Fama and MacBeth 1973). Their conventional test statistics use the time-series standard errors of the slopes without an autocorrelation correction and are treated as exploratory, especially at longer horizons. Because score scales differ across separately trained models, we emphasize coefficient signs and patterns that recur across specifications.

The empirical grid contains related comparisons across input windows, forecast horizons, scaling schemes, model architectures, fusion designs, weighting rules, and sample periods. We do not apply a formal multiple-testing correction. The interpretation therefore emphasizes patterns that recur across specifications and the separate benchmark, transaction-cost, tradability, and extension-sample evidence rather than individual cells.

3. Replicating and extending the representation comparison

Jiang et al. (2023) compare logistic classifiers and CNN1D across alternative scaling schemes and show that image-scaled CNN1D can match the portfolio performance of the chart CNN. We first assess whether this representation result persists through 2024. We then extend the portfolio-level comparison with direct tests of learned-model rank overlap, conditional predictive content, and multimodal complementarity.

3.1. Replication: scaling and temporal sequence models

Table 1 reports annualized gross Sharpe ratios for weekly high-minus-low portfolios over 2001–2024. Panel A reproduces the logistic comparison across cumulative-return, devolatized-return, and image scaling. Panel B reports image-scaled CNN1D and chart-CNN results alongside the multimodal extension. Complete monthly and quarterly results appear in the online appendix.

Table 1: Replication and extension of the weekly representation comparison

	5-day input		20-day input		60-day input	
Model	EW	VW	EW	VW	EW	VW
Panel A. Encoding with a logistic classifier						
Logistic, cumulative scale	0.18	−0.04	−0.41	−0.51	−0.26	−0.53
Logistic, devolatized scale	2.80	0.79	2.73	0.48	2.70	0.66
Logistic, image scale	4.88	1.25	5.22	1.09	5.44	1.12
Panel B. Flexible models with image scale						
CNN1D	6.24	1.16	6.55	1.12	5.89	2.06
Chart CNN	5.88	1.39	6.52	1.44	3.72	1.18
Multimodal	6.01	1.16	6.48	1.52	5.53	1.75

Note. The table reports annualized gross Sharpe ratios of non-overlapping 5-day high-minus-low decile portfolios from 2001 through 2024. Panel A and the CNN1D and chart-CNN rows in Panel B repeat the corresponding model comparisons in Jiang et al. (2023), Table IX; the multimodal row is an extension. EW and VW denote equal- and value-weighted legs. Image scale is the specification label for local high–low scaling of the price and moving-average channels; volume is scaled separately. Chart CNN receives the corresponding binary image. Multimodal is the feature-concatenation model. Each row uses that model's available score cross-section; the rows are not restricted to the exact three-model stock-date intersection used in the overlap tests. The online appendix reports the full scaling and return-horizon grids.

Panel A confirms Jiang et al.'s scaling result in the extended sample. In the representative $I20/R5$ specification, the logistic Sharpe ratio increases from −0.41 under the cumulative-return scale to 5.22 under the image scale, with the devolatized-return scale between them. The same ordering holds across the other input intervals. Local high–low scaling therefore exposes a strong return ranking before temporal learning is introduced.

Local high–low scaling jointly normalizes open, high, low, close, and moving-average values within each lookback window, while volume is scaled separately. The transformation removes nominal price levels and preserves the relative positions of prices, ranges, and the moving average within recent market history. A given price movement occupies a larger share of the normalized range in a narrow-range window and a smaller share in a wide-range window. The locally scaled sequence retains this information as continuous channels without the pixel layout. Its logistic performance shows that this transformation already provides an economically meaningful representation of recent price history.

Panel B confirms the second part of the original representation result. Temporal convolution applied to the locally scaled sequence reproduces the chart CNN's weekly portfolio performance.

In the representative $I20/R5$ specification, the Sharpe ratio increases from 5.22 for the image-scale logistic classifier to 6.55 for CNN1D, which is nearly identical to 6.52 for the chart CNN. CNN1D remains competitive across the other input intervals. Together, Panels A and B replicate the finding that local high–low scaling exposes the baseline ranking and that a temporal model can recover the chart CNN's weekly portfolio performance without the binary pixel layout.

The replicated longer-horizon grid shows an important boundary. Local high–low scaling remains the strongest logistic representation, but temporal convolution adds little once the forecast horizon extends beyond one week. In $I20/R20$, the image-scale logistic classifier reaches a Sharpe ratio of 2.37, compared with 2.15 for CNN1D, and the difference becomes more pronounced as quarterly portfolio performance weakens. Local high–low scaling therefore remains informative across horizons, while the additional contribution of temporal learning is concentrated at the weekly horizon.

Value weighting provides a second boundary on the weekly result. In $I20/R5$, Sharpe ratios near 6.5 for the three equal-weighted learned portfolios fall to between 1.12 and 1.52 under value weighting. The models remain close to one another, preserving the representation result, and the smaller economic magnitude indicates that much of the weekly spread comes from stocks receiving little weight in value-weighted portfolios. The later tradability screens examine this concentration directly.

The representative $I20/R5$ decile profiles show that the weekly performance reflects a broad and similar ranking across the three learned models. Equal-weighted annualized returns rise from approximately -35% to -32% in the lowest-score decile to 53% to 55% in the highest, with generally increasing returns through the middle deciles. The especially steep increase from decile 9 to decile 10 shows additional concentration in the highest-ranked tail. Value-weighted profiles are flatter and receive little contribution from the lowest-score portfolio, reinforcing the smaller-stock boundary identified above.

The replicated portfolio results leave an unresolved question. CNN1D applied to the locally scaled sequence reproduces the chart CNN's weekly performance, but the models retain differences in input precision, filter geometry, and depth that can alter individual scores. Similar Sharpe ratios therefore need not imply that the models rank the same stocks. The first extension examines this distinction directly.

3.2. Extension: representation overlap and conditional content

Jiang et al.'s Table IX compares portfolio performance but does not report direct rank correlations between chart-CNN and CNN1D scores. Table 2 reports average date-level Spearman correlations on the exact common stock-date panel. These statistics measure how similarly the models order stocks within each formation date and show what the replicated portfolio comparison conceals.

Rank overlap is strongest in the weekly specifications where portfolio performance is strongest. For $I5/R5$ and $I20/R5$, pairwise Spearman correlations range from 0.67 to 0.84. The similar weekly portfolio results are therefore accompanied by substantial overlap in the underlying stock rankings.

The 60-day input provides the clearest boundary to the common weekly pattern. In $I60/R5$, the chart CNN/CNN1D rank correlation falls to 0.44 as their Sharpe ratios separate to 3.72 and 5.89. Panel B further shows that CNN1D preserves its weekly ranking more consistently across input windows than the chart CNN. The longer lookback window therefore reveals a representation difference that is less visible with the 5- and 20-day inputs.

Table 2: Rank overlap across learned models and input windows

Specification	Chart/CNN1D	Chart/Multimodal	CNN1D/Multimodal
Panel A. Across-model rank overlap			
$I5/R5$	0.73	0.74	0.84
$I20/R5$	0.67	0.72	0.80
$I60/R5$	0.44	0.65	0.69
$I5/R20$	0.50	0.67	0.71
$I20/R20$	0.53	0.65	0.72
$I60/R20$	0.44	0.55	0.63
$I5/R60$	0.45	0.45	0.68
$I20/R60$	0.44	0.45	0.64
$I60/R60$	0.38	0.46	0.63
Panel B. Within-model rank stability for the 5-day return horizon			
Model	$I5/I20$	$I5/I60$	$I20/I60$
Chart CNN	0.60	0.22	0.34
CNN1D	0.64	0.54	0.71
Multimodal	0.56	0.44	0.59

Note. Panel A reports time-series averages of formation-date Spearman correlations on the common stock-date panel for which all three learned scores and the future return are observed. Panel B compares scores from two input windows within the same model family. All statistics use the 2001–2024 evaluation window.

The rank correlations describe how similarly the models order stocks. Table 3 examines conditional predictive content by regressing future returns jointly on the three model scores. The reported Fama–MacBeth t-statistics show whether each score remains positively related to future returns after conditioning on the other learned-model scores.

Table 3: Joint encompassing regressions for learned-model scores

Specification	Chart CNN	CNN1D	Multimodal	Average R^2 (%)
$I5/R5$	6.95	12.56	8.39	0.70
$I20/R5$	9.51	11.80	11.91	0.80
$I60/R5$	−1.12	18.19	9.95	0.95
$I5/R20$	−0.26	4.39	−0.73	0.52
$I20/R20$	3.93	5.76	3.36	0.65
$I60/R20$	−0.85	4.66	0.28	1.06
$I5/R60$	−0.74	2.16	2.56	0.50
$I20/R60$	−0.50	2.83	1.37	0.75
$I60/R60$	−0.46	3.97	0.71	1.13

Note. The first three columns report the conventional Fama–MacBeth t-statistic for each score in a joint date-level cross-sectional regression over 2001–2024. The statistic uses the time-series standard error $\mathrm{sd}(\beta_t)/\sqrt{T}$ without an autocorrelation correction. It is therefore exploratory, especially at longer horizons and in view of the model search. The last column is the time-series average of the date-level joint-regression R^2.

The regressions sharpen the overlap result. In $I5/R5$ and $I20/R5$, all three scores retain positive conditional associations, showing that substantial rank overlap coexists with predictive variation in each score. In $I60/R5$, the chart statistic becomes weak while CNN1D remains large, consistent with the earlier separation in rankings and portfolio performance. CNN1D is positive throughout the full horizon grid and provides the most consistent conditional evidence.

3.3. *Extension: complementarity across representations*

The second extension tests complementarity directly by combining chart and sequence information. In the central $I20/R5$ specification, the multimodal model reaches a Sharpe ratio of 6.48, compared with 6.52 for the chart CNN and 6.55 for CNN1D. The best single branch also remains strongest in the other weekly specifications. The fusion result therefore points to limited complementarity between the two representations.

The same pattern appears when the branches are combined after separate training. Late fusion averages the chart CNN and CNN1D probabilities or logits and produces weekly performance close to the individual branches. The agreement between feature concatenation and late fusion shows that limited gains persist across fusion designs. Detailed comparisons appear in the online appendix.

Taken together, the multimodal results point to limited complementarity. Fusion preserves the strong weekly portfolio performance across feature-level and prediction-level designs, while the best individual branch retains the highest performance. Combined with the score correlations, this pattern indicates that the chart and sequence representations recover largely overlapping predictive information from the same recent price history. The joint regressions show that some conditional variation remains in the individual scores.

4. Economic value: implementation and fragility

The learned models produce a strong weekly ranking in the unrestricted universe. Whether that ranking represents implementable economic value is less clear. We evaluate it in four steps: comparison with transparent price rules, deduction of a common trading-cost schedule, restriction to more tradable stocks, and performance in the 2020–2024 extension sample. The central distinction is between a strong unrestricted return-ranking result and a portfolio result that is sensitive to weighting, liquidity, and sample period.

4.1. *Gross performance and the cost hurdle*

Table 4 compares the representative $I20/R5$ learned portfolios with MOM, STR, WSTR, and TREND over 2001–2024. Each method retains its own score definition and lookback, but all enter the same weekly non-overlapping decile construction, equal-weighting rule, and transaction-cost adjustment. The rows are therefore comparable as portfolio strategies, although method-specific histories can change the eligible stock-date samples.

Table 4: Weekly equal-weighted learned models and price-based benchmarks

Method	Gross return	Gross SR	Net return	Net SR	Turnover
Chart CNN $I20/R5$	84.9%	6.52	51.3%	3.94	6.64
CNN1D $I20/R5$	89.1%	6.55	56.5%	4.16	6.46
Multimodal $I20/R5$	87.5%	6.48	54.5%	4.04	6.56
MOM	−2.3%	−0.08	−7.1%	−0.25	0.95
STR	44.3%	1.83	27.3%	1.13	3.32
WSTR	59.3%	2.86	24.9%	1.20	6.71
TREND	52.7%	2.49	27.5%	1.31	4.95

Note. Returns and Sharpe ratios are annualized. Turnover is monthly-equivalent conventional half-turnover summed across the two legs. The transaction-cost deduction uses full absolute traded notional. Learned and benchmark strategies draw from the same CRSP source universe, though method-specific lookback requirements can change eligible stock counts.

The learned portfolios clear the gross benchmark hurdle by a wide margin: their Sharpe ratios are near 6.5, compared with 2.86 for WSTR, the strongest transparent rule. Their turnover is also close to WSTR's. Under the common 10–20 basis-point schedule, trading reduces learned-model annualized returns by roughly 33 percentage points, but their larger gross spreads still leave net Sharpe ratios near four; TREND is the strongest net benchmark at 1.31.

This favorable comparison is conditional on equal weighting and a stylized cost schedule. Under value weighting, all three learned portfolios retain positive gross Sharpe ratios, but their net Sharpe ratios are negative in both the baseline and extension samples. The cost schedule also excludes market impact, borrowing costs, short-sale constraints, and capacity. The benchmark table therefore establishes a strong unrestricted ranking, not a claim of scalable profitability. Across the wider grid, the advantage also narrows at longer horizons and when larger stocks receive more weight.

4.2. The tradability boundary

Table 5 locates the unrestricted result within the cross section. We keep the $I20/R5$ scores fixed and re-form the portfolios after price, dollar-volume, and market-capitalization screens. Each filter is applied before sorting, so the exercise asks whether the ranking remains economically useful within a more tradable eligible universe.

Table 5: Tradability and factor diagnostics for representative weekly portfolios

Model	Screen	Net return	Net SR	Carhart α	$t(\alpha)$	Names	Turnover
Chart CNN	Full universe	51.3%	3.94	49.3%	11.81	420	6.65
	Price ≥ \$5	10.1%	1.08	8.1%	3.67	321	6.66
	Dollar volume ≥ \$1m	−10.0%	−0.85	−11.0%	−3.87	241	7.01
	Dollar volume ≥ \$5m	−19.5%	−1.57	−20.6%	−7.27	164	7.07
	No NYSE microcaps	−17.0%	−1.60	−18.1%	−7.38	193	6.89
	Combined	−22.9%	−2.10	−24.0%	−10.21	149	7.02
CNN1D	Full universe	56.5%	4.16	55.2%	12.56	419	6.46
	Price ≥ \$5	13.7%	1.48	12.4%	5.46	319	6.50
	Dollar volume ≥ \$1m	−5.7%	−0.46	−6.2%	−2.21	240	6.68
	Dollar volume ≥ \$5m	−15.0%	−1.17	−15.7%	−5.61	164	6.75
	No NYSE microcaps	−14.8%	−1.37	−15.6%	−6.34	192	6.57
	Combined	−19.0%	−1.68	−19.8%	−8.11	149	6.69
Multimodal	Full universe	54.5%	4.04	53.2%	12.10	419	6.56
	Price ≥ \$5	13.5%	1.52	12.4%	5.46	319	6.63
	Dollar volume ≥ \$1m	−5.2%	−0.42	−5.5%	−1.91	240	6.83
	Dollar volume ≥ \$5m	−14.1%	−1.11	−14.7%	−5.12	164	6.92
	No NYSE microcaps	−14.0%	−1.33	−14.6%	−5.85	192	6.75
	Combined	−19.3%	−1.76	−19.9%	−8.32	149	6.87

Note. The table reports equal-weighted $I20/R5$ portfolios from 2001 through 2024. Names is the average number of stocks in each leg. Combined requires price of at least \$5, dollar volume of at least \$5 million, and market capitalization above the NYSE 20th-percentile breakpoint. Carhart intercepts are annualized and use Newey–West standard errors with four lags.

The result fails quickly as liquidity requirements tighten. A \$5 price screen leaves positive but much smaller net performance; a \$1 million dollar-volume screen makes all three portfolios negative. Excluding NYSE microcaps or combining the screens produces the same conclusion, while turnover remains high. For example, CNN1D moves from a 56.5% net annualized return in the full universe to −5.7% at the \$1 million dollar-volume threshold and −19.0% under the combined screen.

Carhart adjustment does not explain the unrestricted spread: the full-universe annualized alphas remain close to the corresponding net returns. Yet both returns and alphas turn negative under modest dollar-volume and capitalization restrictions. Factor adjustment and implementability thus answer different questions. The ranking is not a conventional factor repackaging, but its measured economic value is concentrated in stocks for which the omitted costs and capacity constraints are likely most severe.

This location is also consistent with the established economics of weekly reversal, whose profits are largest among small and illiquid stocks (Chen et al. 2025). The similarity is descriptive: without an exact common-panel score comparison, the evidence does not establish that the learned models reproduce the reversal signal.

4.3. *Recent-sample fragility*

Table 6 asks whether the historical ranking persists with model parameters, benchmark definitions, and portfolio rules held fixed. It compares the representative learned models and price rules in the 2001–2019 baseline and the 2020–2024 extension. Because the extension spans only five years, its estimates are less precise and should be read as a stability diagnostic.

Table 6: Weekly equal-weighted performance in the baseline and extension periods

	2001–2019				2020–2024			
Method	Gross ret.	Gross SR	Net ret.	Net SR	Gross ret.	Gross SR	Net ret.	Net SR
Chart CNN $I20/R5$	96.3%	7.64	60.0%	4.78	50.7%	3.87	18.5%	1.35
CNN1D $I20/R5$	102.2%	8.09	66.5%	5.28	49.9%	3.34	18.6%	1.18
Multimodal $I20/R5$	97.6%	7.76	64.4%	5.12	49.5%	3.19	17.2%	1.11
MOM	−5.1%	−0.19	−10.0%	−0.36	8.7%	0.26	4.1%	0.12
STR	48.7%	2.04	31.6%	1.33	27.7%	1.10	11.0%	0.44
WSTR	66.2%	3.33	31.7%	1.59	32.9%	1.42	−0.8%	−0.03
TREND	64.4%	3.09	38.3%	1.85	8.1%	0.38	−13.3%	−0.62

Note. Returns and Sharpe ratios are annualized. The learned rows use the representative 20-day input. Benchmark definitions appear in Section 2.4. Portfolio and cost conventions are fixed across periods.

All three learned portfolios weaken sharply after 2019. Their net Sharpe ratios fall from roughly five in the baseline sample to 1.11–1.35 in the extension, although net annualized returns remain positive near 18%. WSTR and TREND fall from net Sharpe ratios above 1.5 to zero or below, and STR also weakens. The decline therefore spans learned representations and transparent short-horizon price rules rather than isolating one architecture.

The appendix shows that 2020 is the chart CNN's low point and that performance partially recovers during 2021–2024. The five-year average consequently combines a sharp initial break with uneven later performance. This pattern is consistent with the recent attenuation reported by Kaczmarek and Pukthuanthong (2025) and with broader evidence of predictor decay (McLean and Pontiff 2016), but it does not distinguish market change from sampling variation or specification search.

Positive equal-weighted extension returns do not overturn the implementation evidence. The representative value-weighted net Sharpe ratios are negative, and the full-sample screens locate the unrestricted returns in difficult-to-trade stocks. The recent sample therefore preserves some return-ranking ability while weakening both its magnitude and its practical interpretation.

5. Conclusion

This study revisits the interpretation of chart-based return prediction. Following Jiang et al. (2023), we separate the local high–low scaling embedded in chart construction from the binary image layout and the nonlinear learning applied afterward. We replicate their central representation comparison through 2024 and extend it with exact score-level overlap tests, joint regressions, fusion designs, tradability screens, and a recent-period evaluation.

The replication confirms that representation choices matter before a flexible model is introduced. Local high–low scaling turns the logistic classifier from weak to strongly predictive, and CNN1D applied to the same continuous, locally scaled sequence closely matches the chart CNN in the strongest weekly specifications. The additional contribution of temporal convolution is concentrated at the weekly horizon. These results confirm Jiang et al.'s conclusion that local high–low scaling is a central source of chart-CNN performance and that the binary image is not necessary to reproduce its weekly portfolio results.

The extensions qualify what similar portfolio performance implies. Chart CNN, CNN1D, and multimodal scores rank many of the same stocks, especially in the strongest weekly specifications, but their rankings are not identical and each score retains conditional variation in parts of the grid. Neither feature-level nor prediction-level fusion consistently improves on the strongest individual branch. The representations therefore recover largely shared information, with limited complementarity at the portfolio level.

The economic evidence draws a sharper boundary. Learned portfolios outperform transparent price rules in the unrestricted equal-weighted universe and remain positive under the common transaction-cost schedule. That result does not survive stronger implementation requirements: net performance is negative under value weighting and after modest dollar-volume or market-capitalization screens. Conventional factor adjustment does not absorb the unrestricted spread, but the returns are concentrated where market impact, borrowing constraints, and capacity are most concerning. All learned models also weaken after 2019, alongside short-horizon price-based benchmarks, although the five-year extension is too short to identify the source of the decline.

Taken together, the evidence supports a historically strong short-horizon ranking more clearly than a scalable trading opportunity. Local high–low scaling recovers much of the representation result, temporal learning strengthens the weekly ranking, and the binary chart adds little incremental portfolio performance. The score-level tests expose this shared content directly, while recent performance and tradability define its narrow economic scope.

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